1	90

Series Expansions You Should Know

$$
\begin{aligned}
e^x &= 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots &&= \sum_{n=0}^{\infty} \frac{x^n}{n!} \\
\cos x &= 1 - \frac{x^2}{2} + \frac{x^4}{4!} - \dots &&= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \\
\sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots &&= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \\
\cosh x &= 1 + \frac{x^2}{2} + \frac{x^4}{4!} + \dots &&= \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!} \\
\sinh x &= x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots &&= \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!} \\
\frac{1}{1-x} &= 1 + x + x^2 + x^3 + \dots &&= \sum_{n=0}^{\infty} x^n \\
\frac{1}{1+x} &= 1 - x + x^2 - x^3 + \dots &&= \sum_{n=0}^{\infty} (-x)^n \\
\tan^{-1} x &= x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots &&= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \\
\ln(1+x) &= x - \frac{x^2}{2} + \frac{x^3}{3} - \dots &&= \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}
\end{aligned}
$$